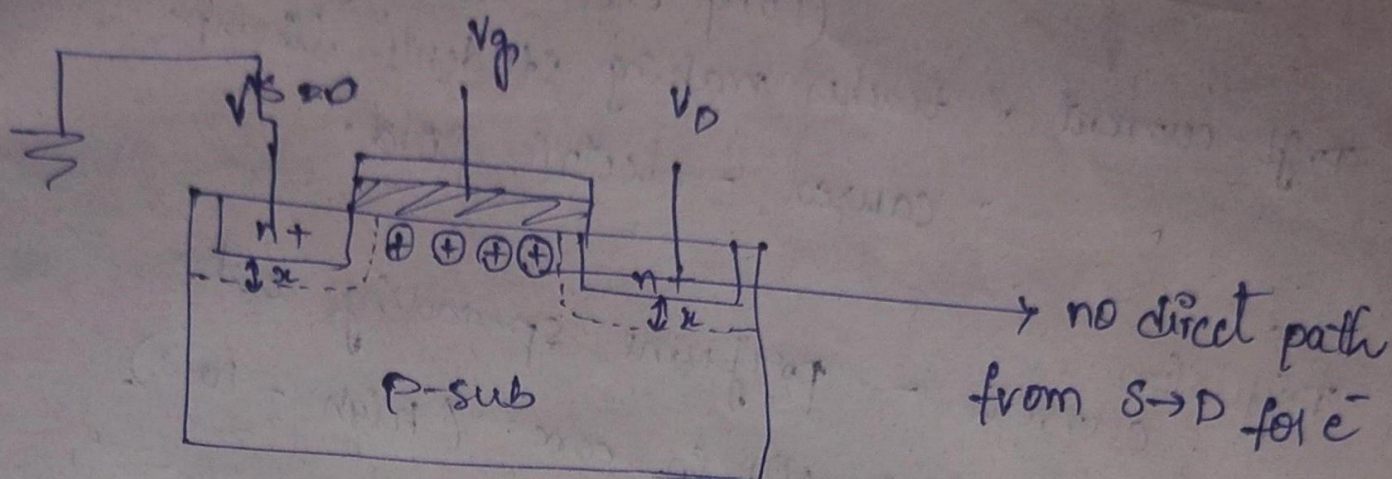


* cutoff mode of operation *



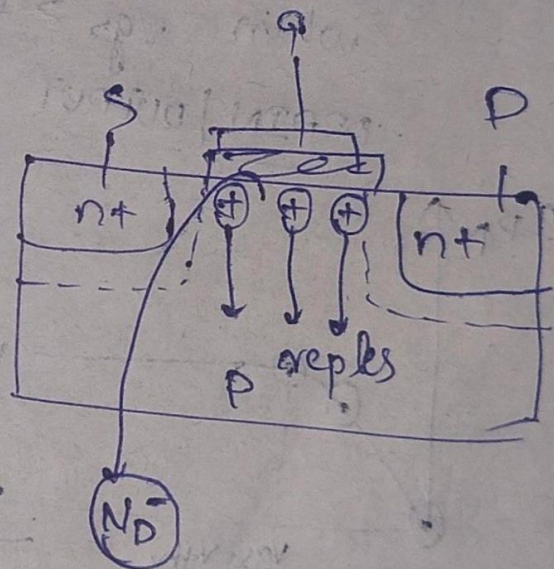
No bias $\rightarrow$ depletion region
 x length = fix

① $V_g = 0$ cutoff $V_g = V_s = V_d = 0$
 no channel formation $V_{gs} = 0$

② $V_d = 0$
 $V_g = 0$
 $V_b = 0$
 $V_g \uparrow$

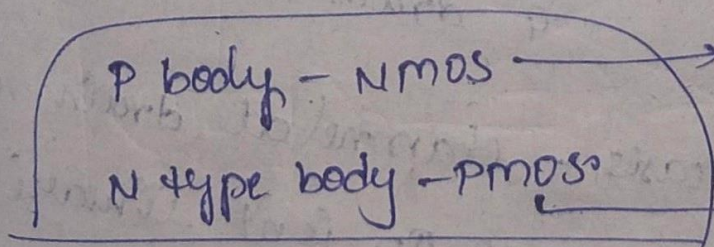
$\oplus \oplus \oplus$ repels

$g \rightarrow s$ (fixed bias).



NMOS gate +ve

PMOS gate -ve
 drain $\rightarrow$



high power, low noise consumption
 $V_g = +ve$

conducts when $V_g = -ve$
 low power consumption
 high noise immunity

IV Characteristics (Enhancement type MOSFET)

(Long channel)

Drift current - teacher making students line up
- caused - electric field.

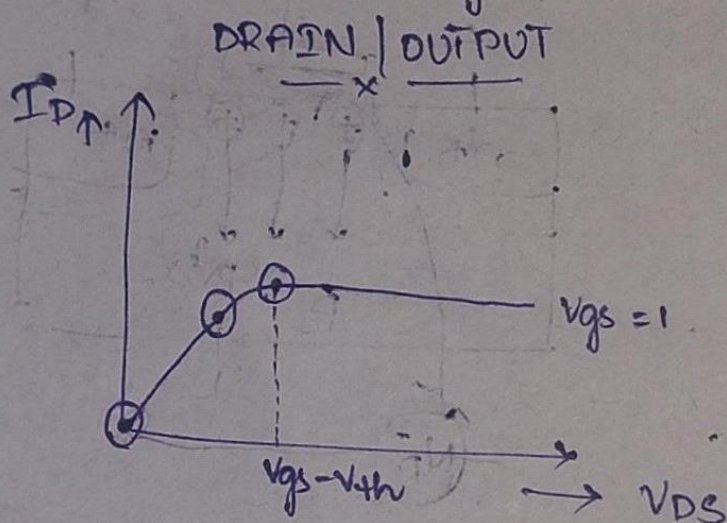
Diffusion current - perfume spreading out
- diff in conc (high - low).

I_D vs V_{DS} ($V_{GS} = \text{const}$).

① cutoff region - $V_{GS} < V_{th}$

$$V_{DS} = 0$$

when $V_{GS} > V_{th} \rightarrow$ channel forms. & $I_D = 0$



This drain characteristic graph because it shows how the current at drain is affected by V_{DS} .

② as V_{DS} increases,
 I_D flows linearly

channel is more at source
& drain is less

③ as V_{DS} further increases - channel at drain reduces
& source channel increases - so I_D isn't linearly.

④ V_{DS} is further increases the channel is like a
point near drain.

$$V_D = V_{DSat}$$

pinch off point

⑥ even on increasing V_{DS} pinch off voltage starts moving towards source

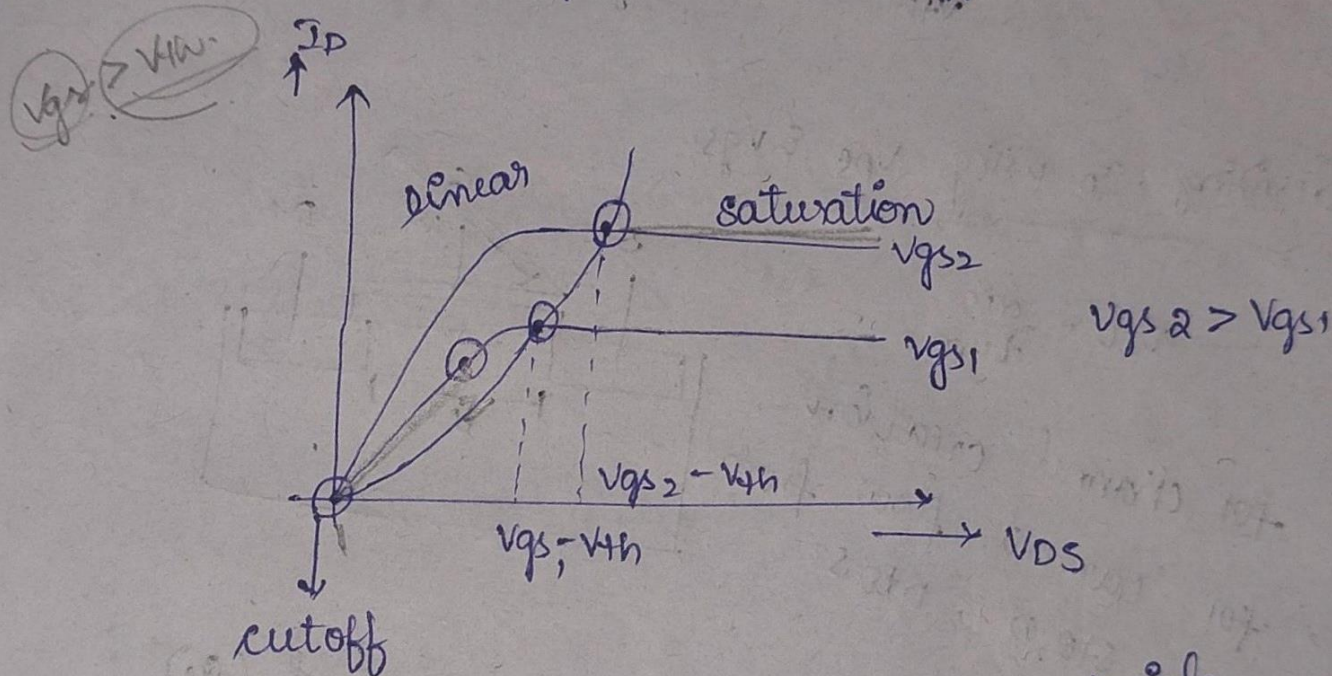
* $V_{GS} < V_{th}$ - No channel

* To form channel $V_{GS} > V_{th}$ (shld be) so in the

$V_{GS} > V_{th}$ channel forms but no I_D so

this channel is formed due to $-V_{th} + V_{GS}$ voltage

* At $V_{DS} = V_{GS} - V_{th}$ pinch off point is formed.



PMOS

$V_{GS} < V_{th} = \text{cutoff}$

$I_D = 0$

deep triode

$V_{GS} > V_{th}$

$V_{DS} \ll V_{GS} - V_{th}$

break down voltage

$V_{DS} \leq V_{GS} - V_{th}$; $V_{GS} > V_{th}$; I_D 's present

triode/linear $\longrightarrow$ acts like resistor.

$V_{DS} \geq V_{GS} - V_{th}$; $V_{GS} > V_{th}$

saturation

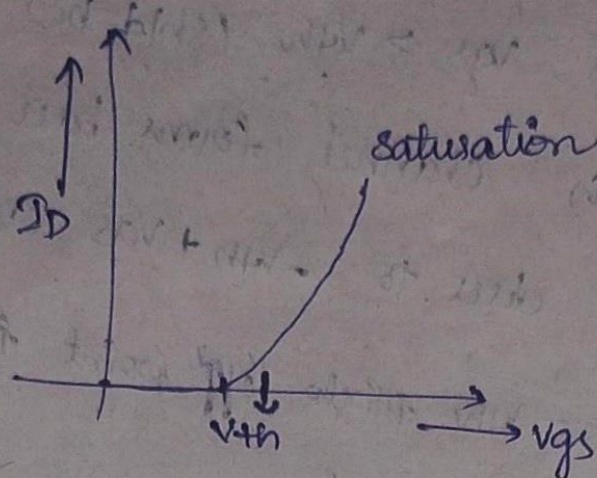
I_D is const wrt V_{DS} .

The deep triode region is a small part of triode, very close to y-axis where V_{DS} is extremely small.

mos acts as near-ideal resistor.

I_D vs V_{gs} (transfer)

It's called transfer because it's where I_D transfer op relation.



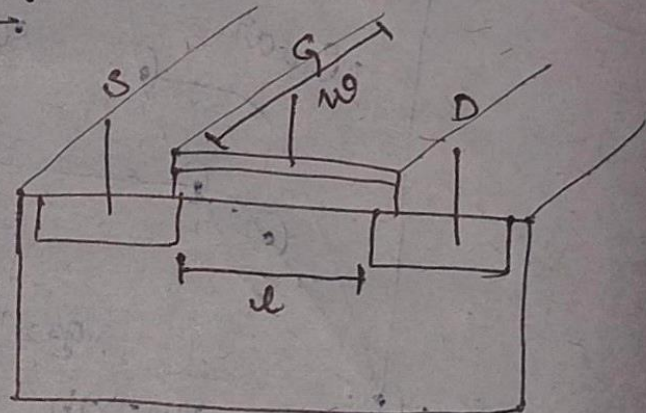
pullup - PMOS - V_{DD}

pull down - NMOS - 0

Deriving I_D using V_{DS} & V_{GS} .

$$v_{el} = \frac{dl}{t_{ym}}$$

V_{GS} = for channel creation
 V_{DS} = for current flow from S to D, D to S



$$\Rightarrow I_{DS} = \frac{\text{charge produced in channel } (Q_c)}{\text{electron transit time } (\tau_{sd})}$$

$$\tau_{sd} = \frac{\text{length } (l)}{\text{velocity of } e^- (v)}$$

time by e^- to S $\rightarrow$ D

where

$$\text{velocity of } e^- (v) = \mu E_{ds}$$

$v \propto E_{ds} \rightarrow$ applied electric field b/w ds

$$E_{ds} = \frac{V_{ds}}{l}, \quad \text{Cof } E = v/l$$

$$\Rightarrow \tau_{sd} = \frac{l}{\mu E_{ds}} = \frac{l}{\mu \left(\frac{V_{ds}}{l} \right)} = \frac{l^2}{\mu V_{ds}}$$

Now,

$$Q_c = \frac{A \epsilon}{d} \times V$$

$$Q = CV$$

$$\Rightarrow C = \frac{A \epsilon}{d}$$

$$\Rightarrow Q_c = \frac{LW \epsilon \epsilon_0}{t_{ox}} \times V$$

Now $V = V_{ds} < V_{gs} - V_{th}$ voltage at gate

effective gate voltage

$$\text{so } V = V_{gs} - V_{th} - \frac{V_{ds}}{2} \rightarrow \text{avg } V_{DS} \text{ in non-sat}$$

so,

$$Q_c = \frac{WL \epsilon \epsilon_0}{t_{ox}} \left((V_{gs} - V_{th}) - \frac{V_{ds}}{2} \right) \quad (2)$$

$$\Rightarrow I_{ds} = \frac{Q_c}{T_{sd}} = \frac{WL \epsilon \epsilon_0}{t_{ox}} \left((V_{gs} - V_{th}) - \frac{V_{ds}}{2} \right) \times \frac{L^2}{\mu V_{ds}}$$

put $d = t_{ox}$

$$I_{ds} = \frac{\epsilon \epsilon_0 W \mu}{d L} \left[(V_{gs} - V_{th}) - \frac{V_{ds}}{2} \right] V_{ds}$$

$$I_{ds} = C_{ox} \frac{\mu}{L} \left[(V_{gs} - V_{th}) - \frac{V_{ds}}{2} \right] V_{ds} \quad (3)$$

$$I_{ds} = C_{ox} \frac{\mu}{L} \left[V_{ds} (V_{gs} - V_{th}) - \frac{V_{ds}^2}{2} \right] \quad (4)$$

linear region

now in $V_{ds} = V_{gs} - V_{th}$ for saturation in (3)

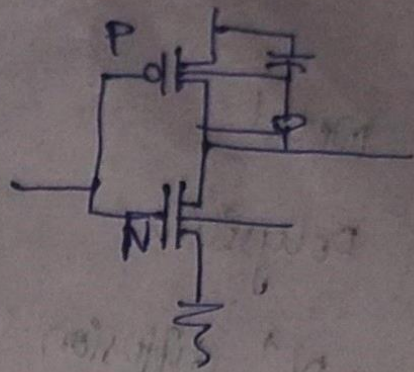
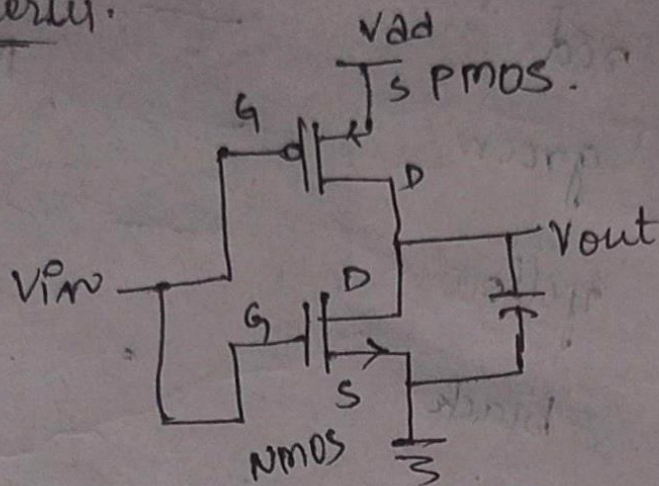
$$I_{ds} = C_{ox} \frac{\mu}{L} \left[(V_{gs} - V_{th}) (V_{gs} - V_{th}) - \frac{(V_{gs} - V_{th})^2}{2} \right]$$

$$I_{ds} = C_{ox} \frac{\mu}{L} \left[\frac{(V_{gs} - V_{th})^2}{2} \right] \rightarrow \text{saturation region}$$

DC Transfer characteristics.

aspect ratio $\frac{W}{L}$

Inverter.



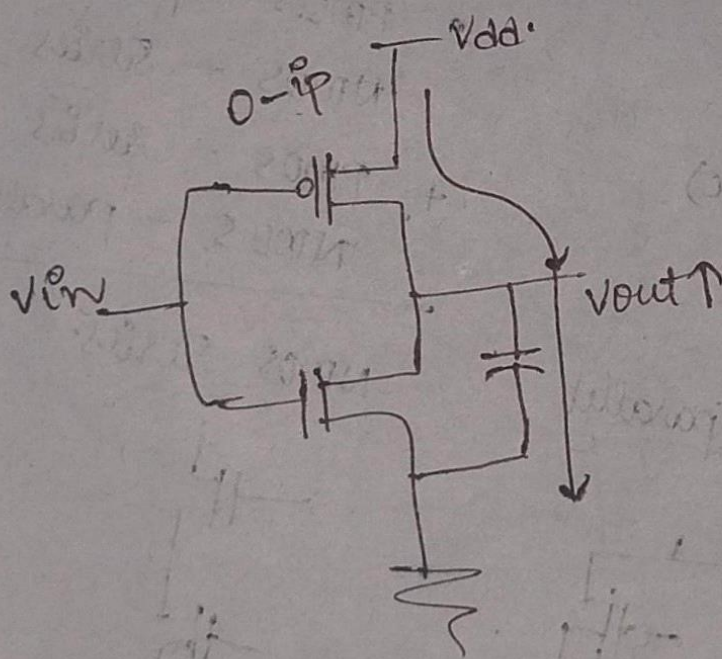
NMOS source - ve potential

PMOS source - +ve potential

NMOS gate $V_{gs} < V_{th}$ - off.
 $V_{gs} > V_{th}$ - ON

PMOS on $V_{gs} < V_{th}$ - OFF ON
 $V_{gs} > V_{th}$ - off.

When $i_p = 0 < V_{th}$ $\rightarrow$ PMOS on, NMOS off



$i_p = 0$ PMOS ON
 $i_p = 1$ NMOS ON.
 Cap is charging so,
 $V_{out} > 1$.

When $i_p = 1$, NMOS is on
 PMOS is off

no direct connection b/w V_{dd} .

capacitor drops down $\rightarrow$ ground.
 $V_{out} = 0$

7/1/25

NMOS

cutoff

linear

saturation

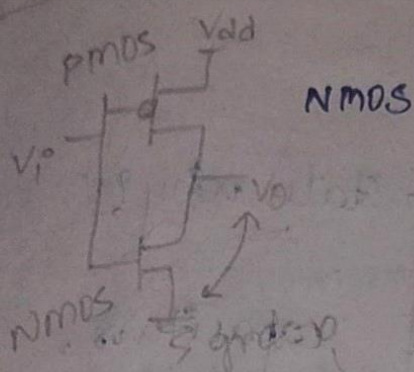
$$V_{gs} < V_{th}$$

$$V_{gs} > V_{th}$$

$$V_{gs} > V_{th}$$

$$V_{ds} < V_{gs} - V_{th}$$

$$V_{ds} > V_{gs} - V_{th}$$



$$V_{in} < V_{th}$$

$$V_{in} > V_{th}$$

$$V_{in} > V_{th}$$

$$V_{out} < V_{in} - V_{th}$$

$$V_{out} > V_{in} - V_{th}$$

PMOS

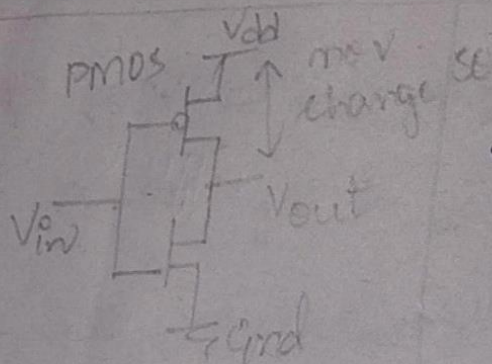
$$V_{gs} > V_{th}$$

$$V_{gs} < V_{th}$$

$$V_{gs} < V_{th}$$

$$V_{ds} > V_{gs} - V_{th}$$

$$V_{ds} < V_{gs} - V_{th}$$



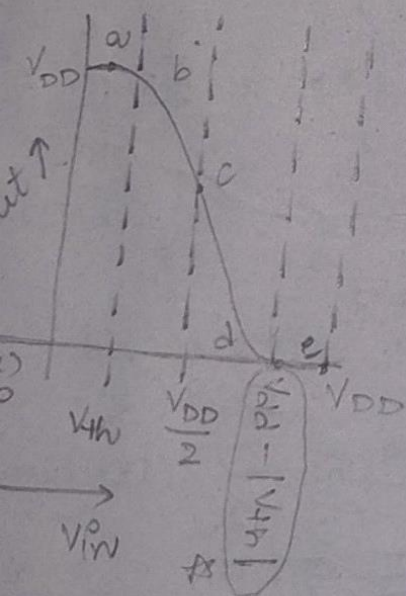
$$V_{in} > V_{th} + V_{dd}$$

$$V_{in} < V_{th} + V_{dd}$$

$$V_{in} < V_{th} + V_{dd}$$

$$V_{out} > V_{in} - V_{th}$$

$$V_{out} < V_{in} - V_{th}$$



Region	PMOS	NMOS
A	linear	cutoff
B	linear	sat
C	sat	sat
D	sat	linear
E	cutoff	linear

Stick diagrams

Metal
 polysilicon
 S, D parts
 N⁺ diffusion
 P⁺ diffusion
 contact

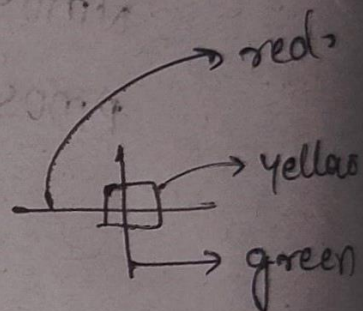
Blue
 red
 green
 yellow
 black

PMOS - S, D - p
b - n

NMOS - S, D - n
b - p

demarkation layer brown

NMOS with depletion
 red over green
 with yellow box
 at center



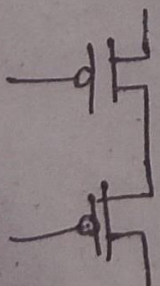
Boolean → CMOS Schematics

$$Y = A \cdot D + A \cdot (B + C)$$

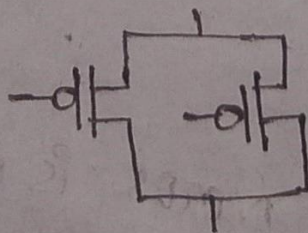
1) pull up PMOS (B+C)

- PMOS - parallel
- NMOS - series
- + PMOS - series
- NMOS - parallel

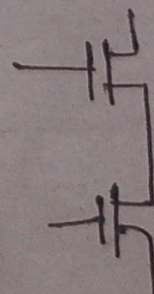
PMOS Series



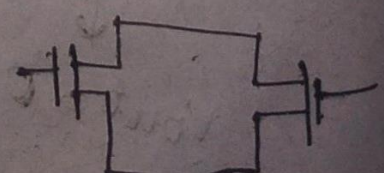
PMOS parallel



NMOS Series



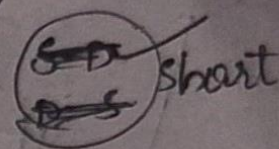
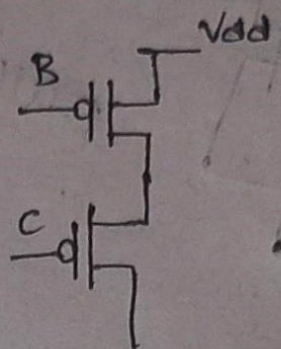
NMOS parallel



$$Y = D + A \cdot (B + C)$$

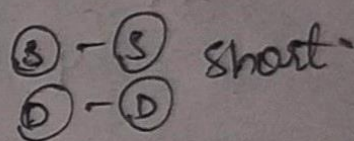
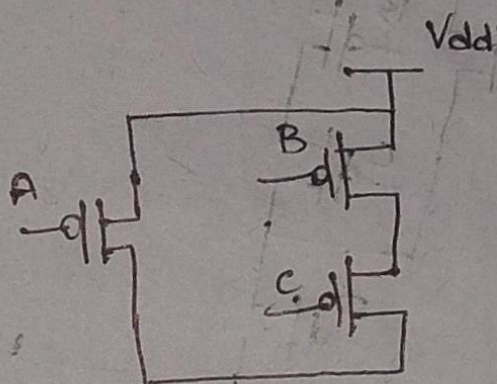
① $B + C$

PMOS - series (+)



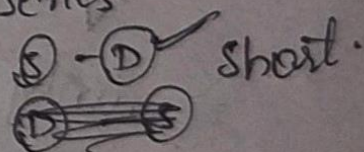
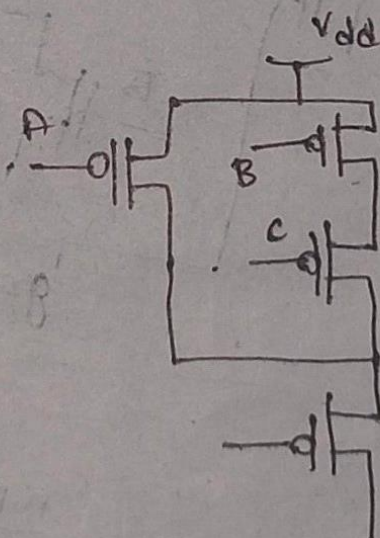
② $A \cdot (B + C)$

PMOS - parallel



③ $D + A(B + C)$

(+) PMOS - Series

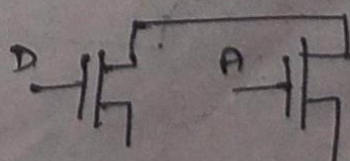
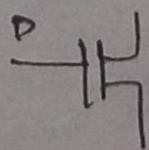


pull up.

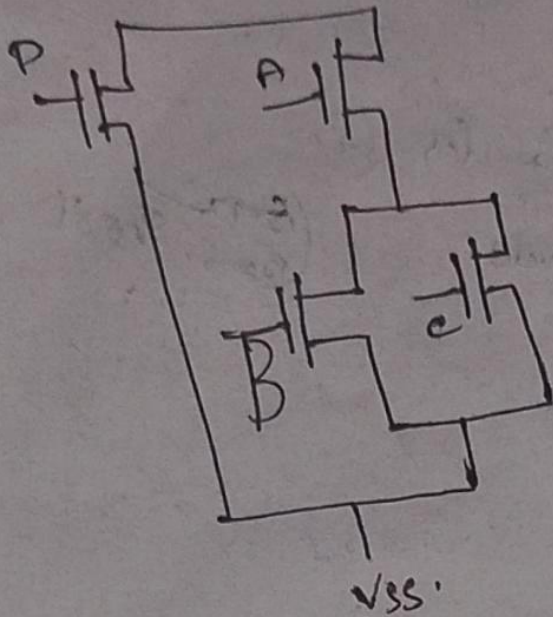
④ D

⑤ $D + A$

NMOS - parallel



⑥ $D + A(B + C)$

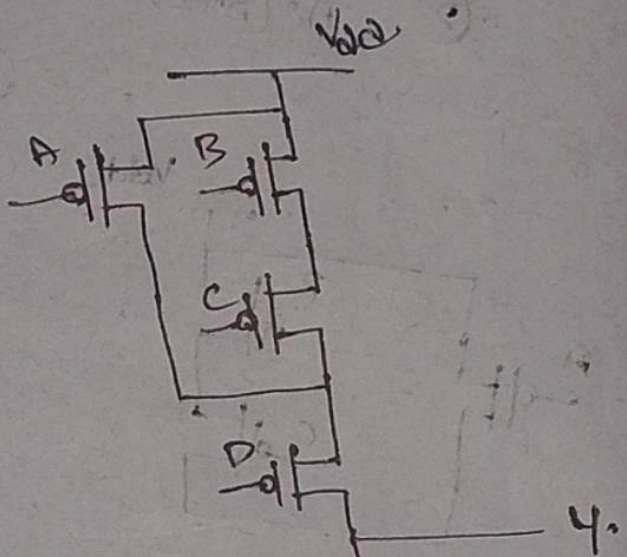


$(B+C)$
 $(+)$ NMOS parallel

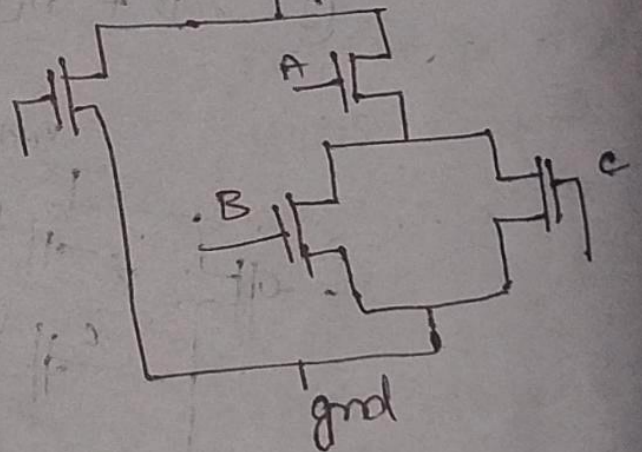
pull down

so,

$$Y = D + A(B+C)$$



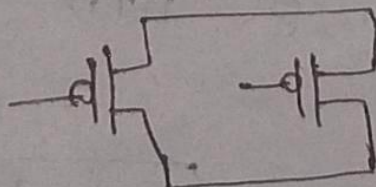
$V = \bar{Y}$
 write same Y place
 degpara oka NOT connect
 cheyali



$$Y = A(B+C) + DE$$

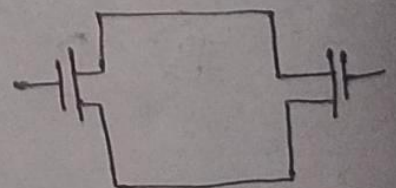
PMOS

$(B+C)$



$A(B+C)$

NMOS



Slick diagrams

$$Y = \overline{A}B + A\overline{B} = \overline{AB + \overline{A}\overline{B}}$$

- PMOS - parallel
- NMOS - series
- + PMOS - series
- NMOS - parallel

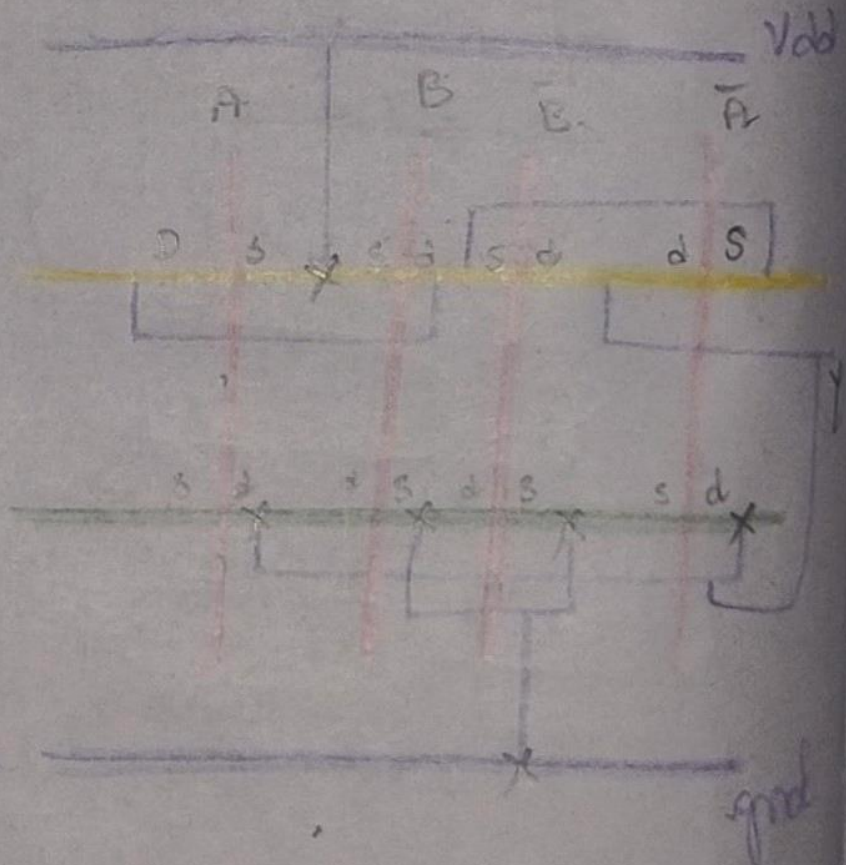
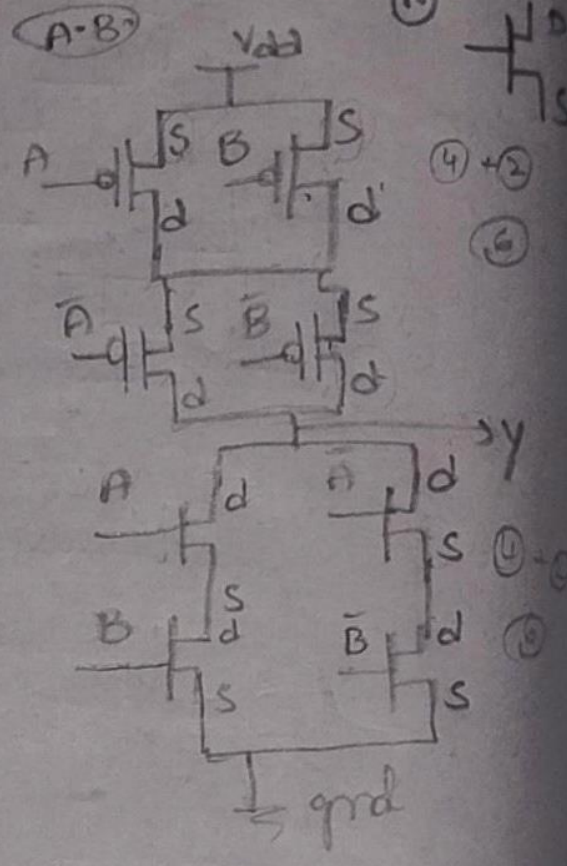
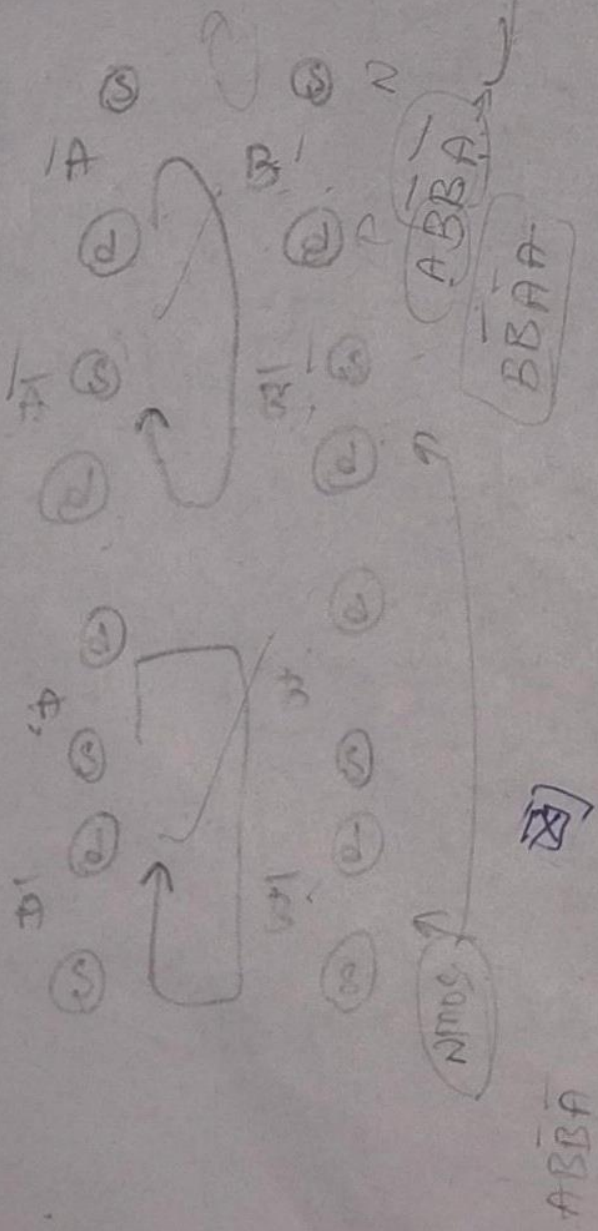
EXOR

A	B	Y	Y'	Z
0	0	0	1	0
0	1	1	0	1
1	0	1	0	1
1	1	0	1	0

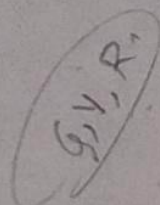
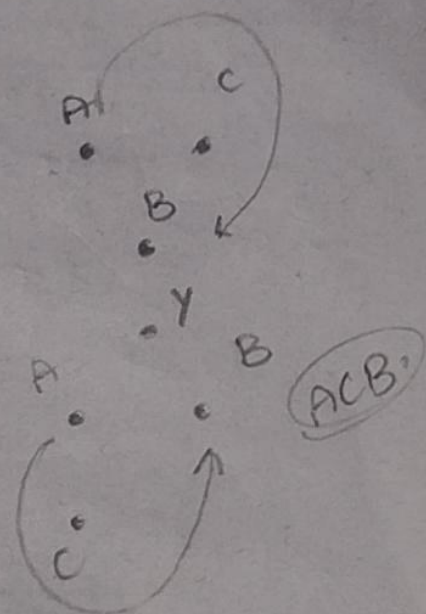
$$Z = \overline{AB + \overline{A}\overline{B}} = \overline{AB + \overline{A}\overline{B}} = (A+B) \cdot A$$

$$Z = Y = AB + \overline{A}\overline{B}$$

ABBA

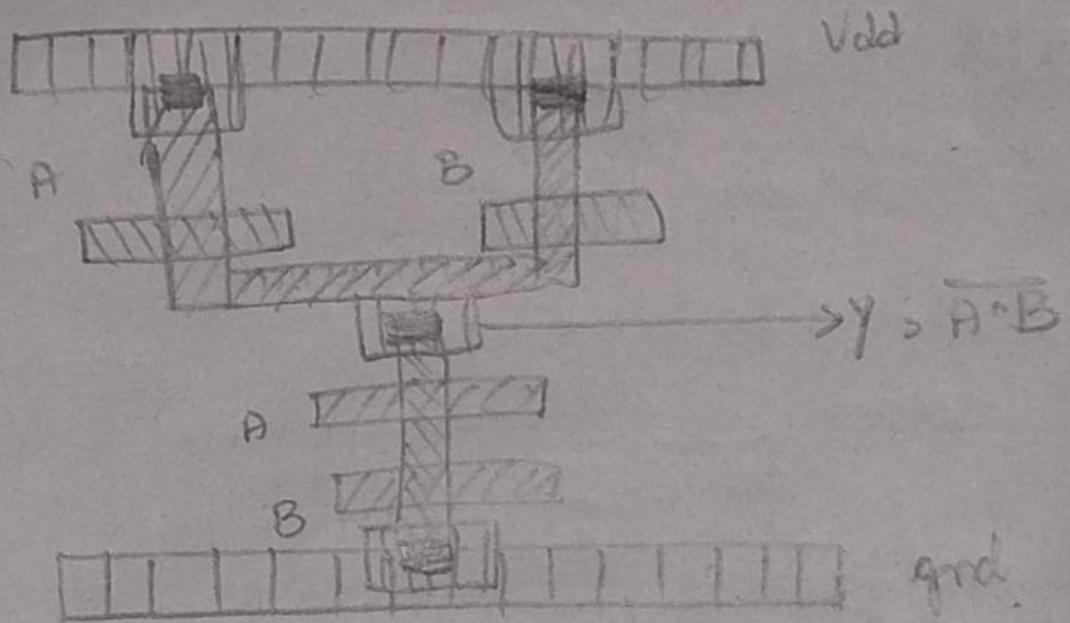
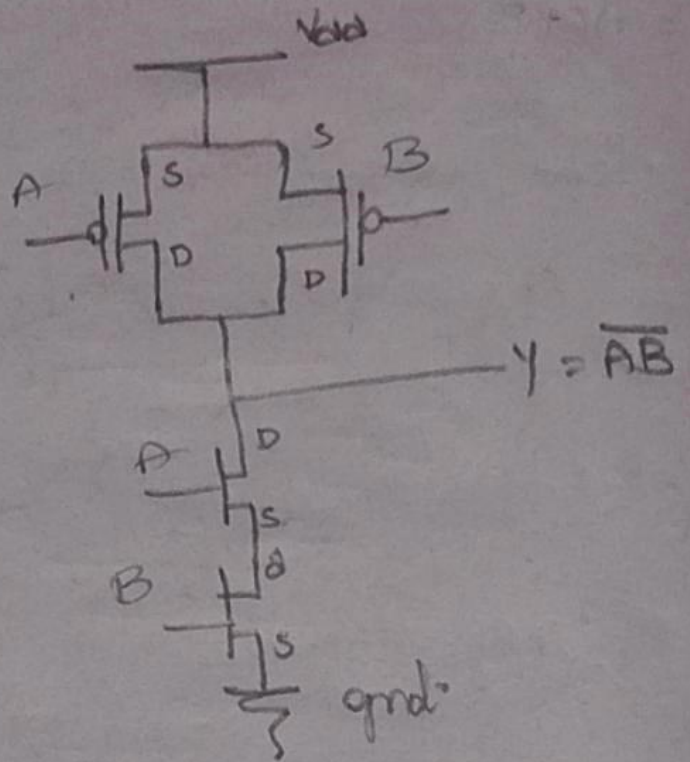
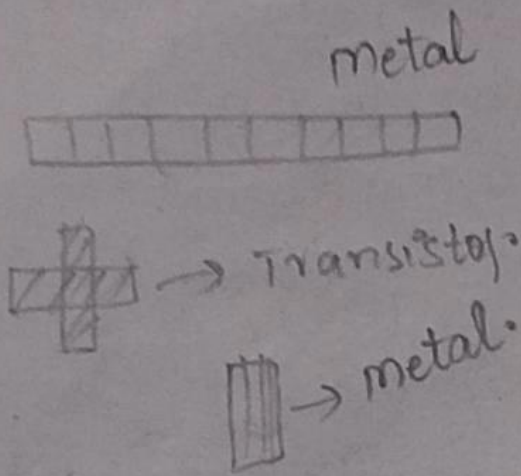


ACB



Layouts.

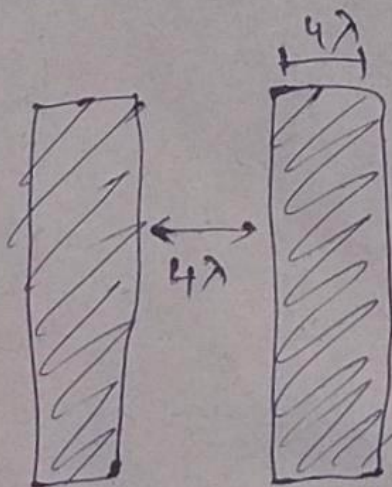
$$Y = \overline{A \cdot B}$$



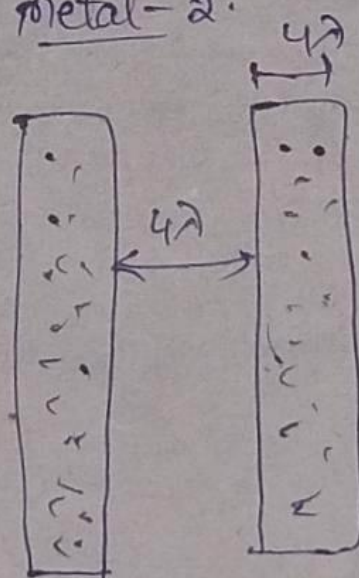
Lambda Based rule.

It prevents closed, flipped areas.

metal-1.

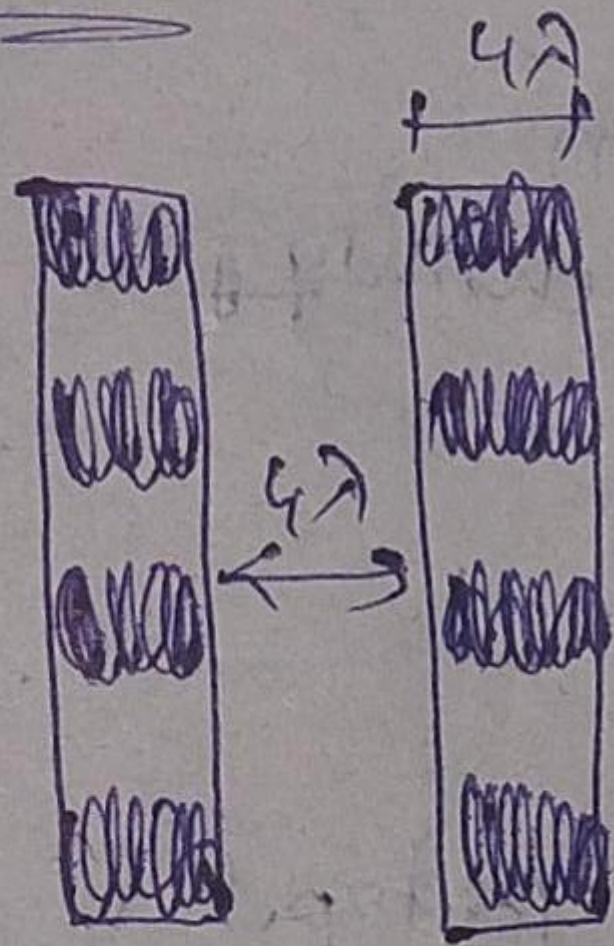


$\overline{AB} + \text{metal-2.}$

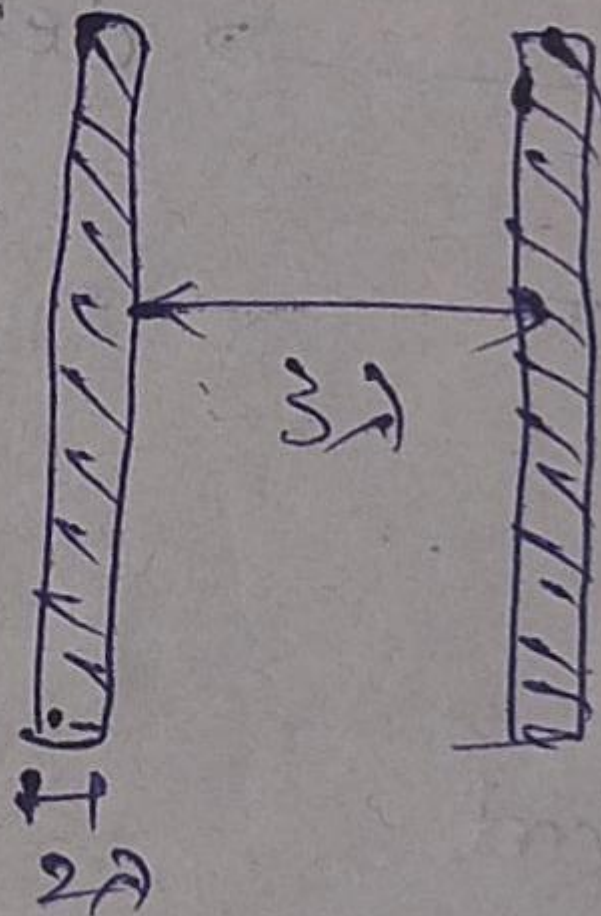


$$\overline{AB} + \text{NOT} = \overline{AB}$$

diffusion



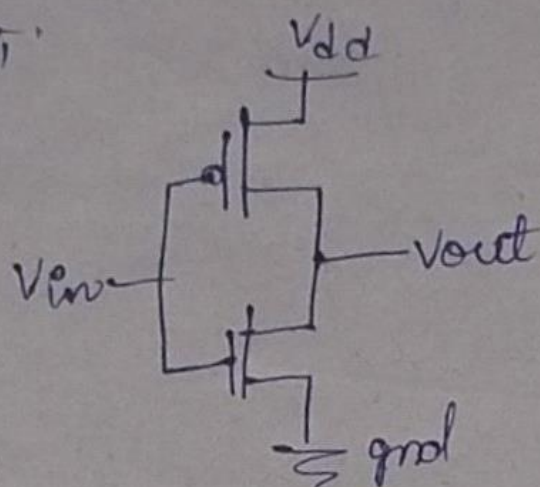
poly silicon



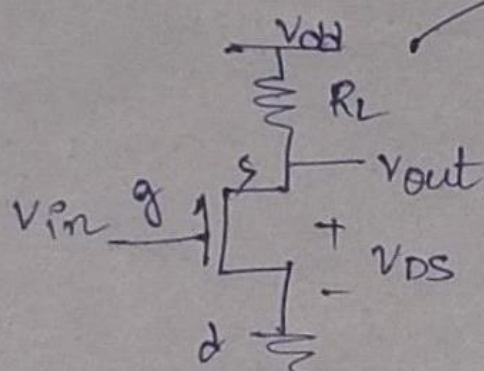
11/2/25

LOADS IN MOSFET

CMOS - NOT



Resistive load.



after PMOS replacing with also it's as NOT only.

$$V_{in} = V_{DD} \text{ ON}$$

$$V_{out} = 0 \text{ (ideal)}$$

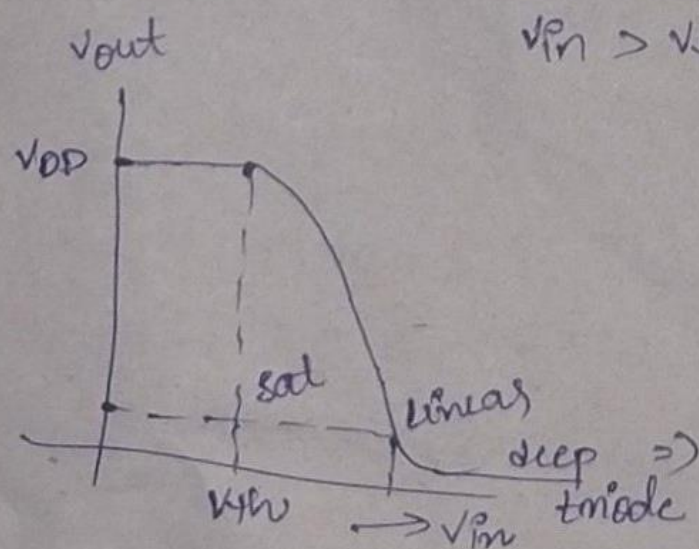
$$\Rightarrow V_{out} = V_{DD} - I_D R_L$$

$$V_{in} = 0 \text{ OFF}$$

$$\text{below } V_{th} \quad V_{out} = V_{DD} \quad \text{so } op = ip$$

① $V_{in} < V_{th}$ - cut off

$V_{in} > V_{th}$ - linear / saturation



$$V_{DS} \geq V_{GS} - V_{th}$$

$$\Rightarrow V_{out} \geq V_{in} - V_{th}$$

considers $V_{GS} = \Delta V$

$$V_{DD} - \Delta V \geq \Delta V + V_{th} - V_{th}$$

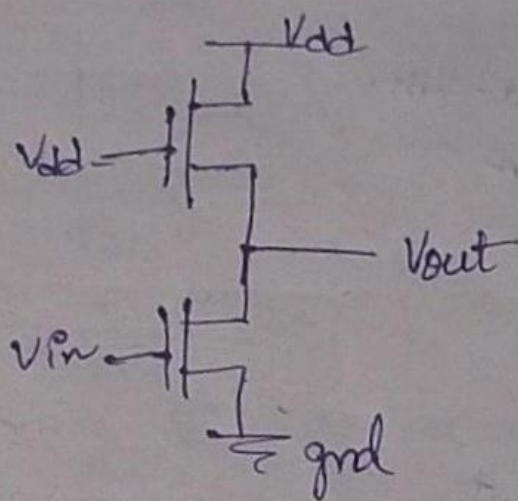
on comparison we say

it's in sat.

limitations - more space on chip, high power consumption

Enhancement load.

PMOS $\rightarrow$ NMOS



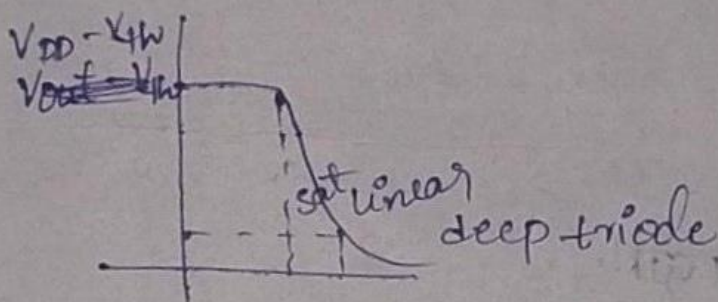
$$V_{in} = V_{DD} \text{ ON}$$

$$V_{out} = 0$$

$$V_{in} = 0 \text{ OFF}$$

$$V_{out} = V_{DD}$$

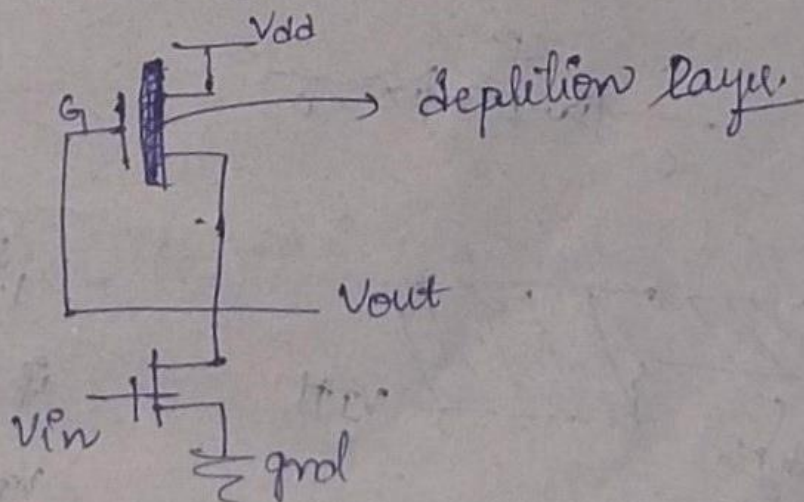
$$\Rightarrow V_{out} = V_{DD} - V_{th}$$



Limitation $\rightarrow$ AS NMOS is always on noise margin is less.

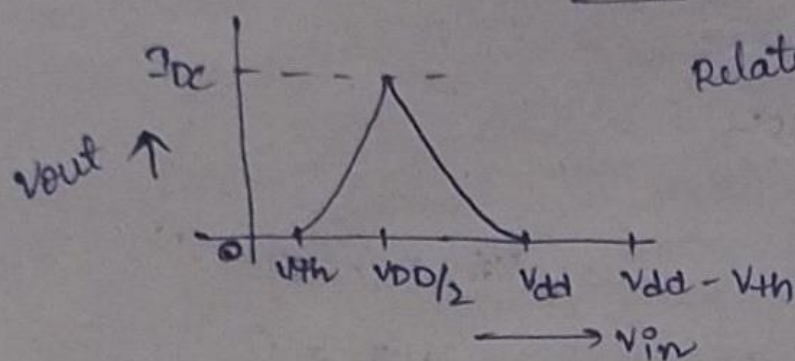
Depletion load.

Irrespective of V_{gs} layer is always on so noise margin is high but high power req.

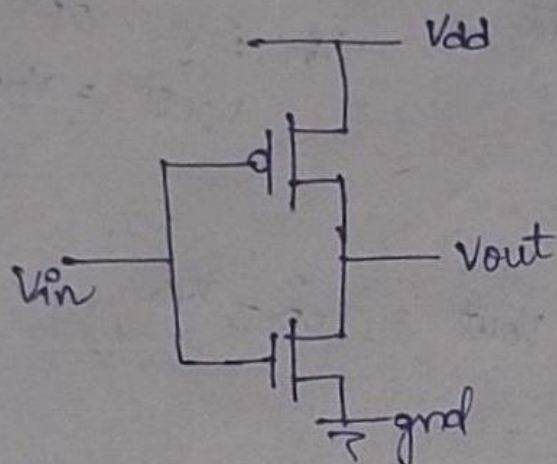


Depletion layer is already put by us.

PMOS Load.



Relation b/w
dc current & voltage

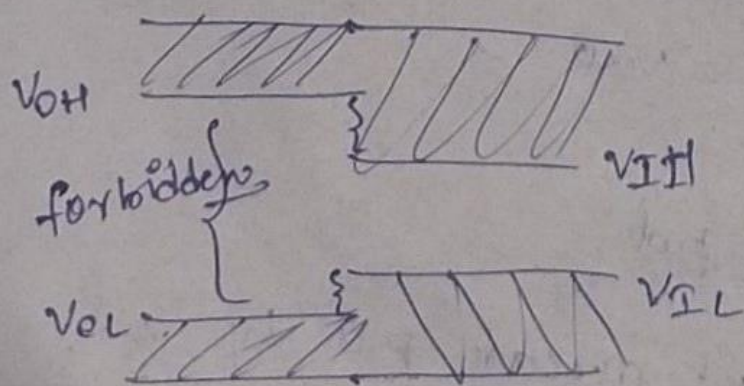
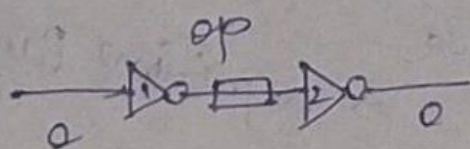


NOISE MARGIN.

V_{IH} - min ip " " " " 1
 V_{OH} - min op voltage is considered as 1.

V_{OL} - max " " " " 0
 V_{IL} " " " " 0

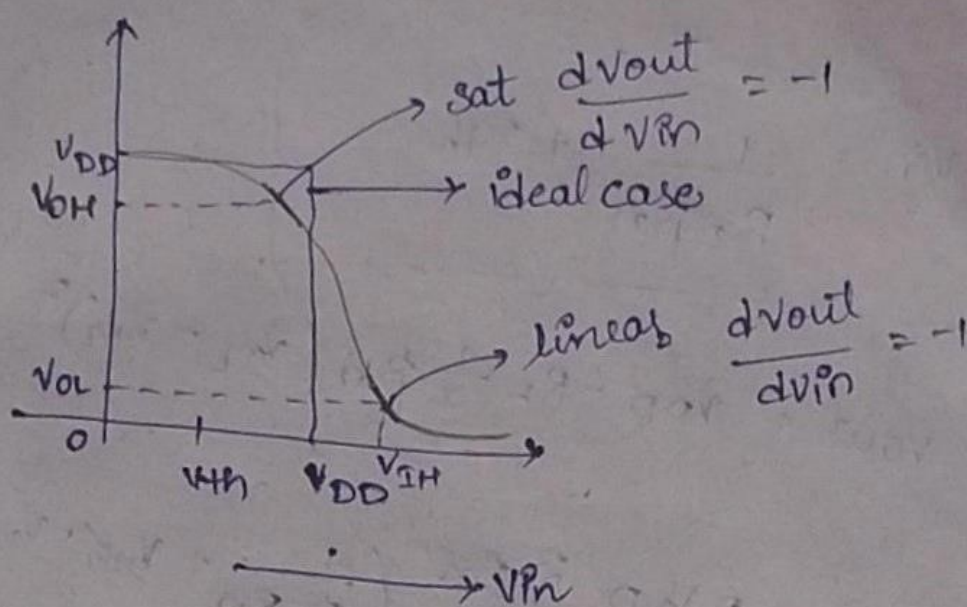
Immunity of gate to resist noise - noise margin



1st op $\rightarrow$ ip 2nd
 if that is b/w the
 $V_{IH} - V_{OH}$ then op
 not affected else
 effected

$$NM_2 = V_{IL} - V_{OL}$$

$$NM_H = V_{OH} - V_{IH}$$

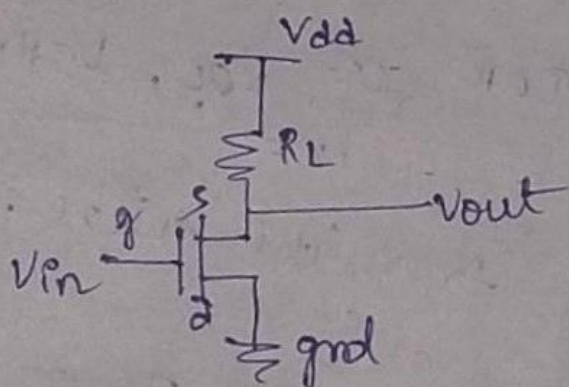


ideally it shd take all the current slowly & shd become V_{DD} .

Consider.

Resistive load

CALCULATION OF V_{OH} , V_{IL}



WKT,

$$V_{out} = V_{DD} - I_D R_L$$

KCL at op

$$I_R = I_D$$

Ohm's law

$$I_R = \frac{V_{DD} - V_{out}}{R_L}$$

$$\Rightarrow \frac{V_{DD} - V_{out}}{R_L} = \frac{1}{2} \beta_{n1} (V_{in} - V_{th})^2$$

diff wrt V_{in} → sat

$$\frac{d}{dV_{in}} \left(\frac{V_{DD} - V_{out}}{R_L} \right)$$

$$\frac{-dV_{out}}{dV_{in}} \cdot \frac{1}{R_L}$$

$$= \beta_{n1} (V_{in} - V_{th})$$

$$\Rightarrow \frac{1}{R_L \beta_{n1}} = V_{in} - V_{th}$$

$$V_{IL} = V_{in} = \frac{1}{R_L \beta_n} + V_{th}$$

$$V_{OH} = V_{out} = V_{DD} - R_L \frac{1}{2} \beta_n (V_{IL} - V_{th})^2$$

$$= V_{DD} - \frac{R_L \beta_n}{2} \left(\frac{1}{R_L \beta_n} + V_{th} - V_{th} \right)^2$$

$$V_{OH} = V_{DD} - \frac{1}{2 R_L \beta_n}$$

CALCULATION OF V_{OL} , V_{IH}

this is where transistor enters into saturation

$$V_{out} = V_{DD} - I_D R_L$$

KCL at op $I_R = I_D$

$$I_D = \frac{V_{DD} - V_{out}}{R_L}$$

$$\frac{V_{DD} - V_{out}}{R_L} = \beta_n \left((V_{in} - V_{th}) V_{out} - \frac{V_{out}^2}{2} \right)$$

diff wrt V_{in}

$$\frac{d}{dV_{in}} (V_{in} - V_{th}) V_{out}$$

$$\frac{1}{R_L \beta_n}$$

$$= -V_{in} + V_{th} + V_{out} + V_{out}$$

$$V_{in} = 2V_{out} + V_{th} - \frac{1}{R_L \beta_n} \left(\frac{1}{2} \right)$$

$$V_{out} = V_{DD} - R_L \beta_n \left(2V_{out} + V_{th} - \frac{1}{R_L \beta_n} - V_{th} \right)^2$$

$$V_{out} = V_{DD} - R_L \beta_n \left((2V_{out} + V_{th}) \frac{1}{R_L \beta_n} - V_{th} \right) V_{out} - \frac{V_{out}^2}{2}$$

$$V_{out} = V_{DD} - R_L \beta_n \left(2V_{out}^2 - V_{out} - \frac{V_{out}^2}{2} \right)$$

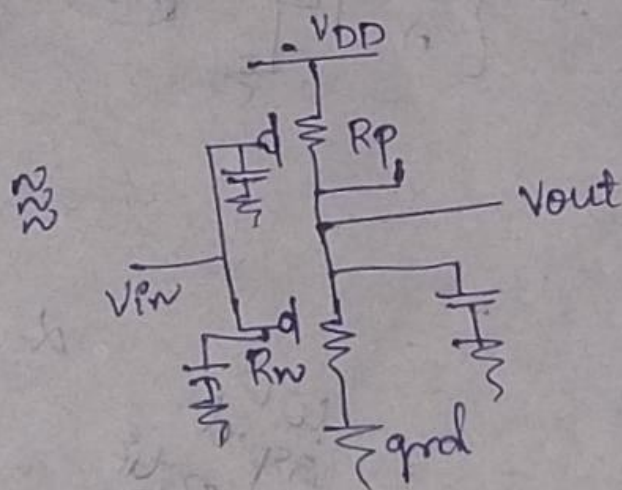
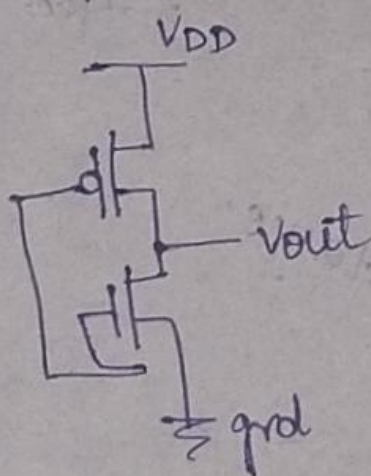
$$V_{out} = V_{DD} - \frac{3}{2} R_L \beta_n V_{out}^2 + V_{out}$$

$$V_{OL} = V_{out} = \sqrt{\frac{2}{3} \frac{V_{DD}}{R_L \beta_n}}$$

$$V_{in} = \sqrt{\frac{8}{3} \frac{V_{DD}}{R_L \beta_n}} + V_{th} - \frac{1}{\beta_n R_L} = V_{THL}$$

RC MODEL

source C doesn't effect transient delay



R_p, R_n = Resistor of N, PMOS.

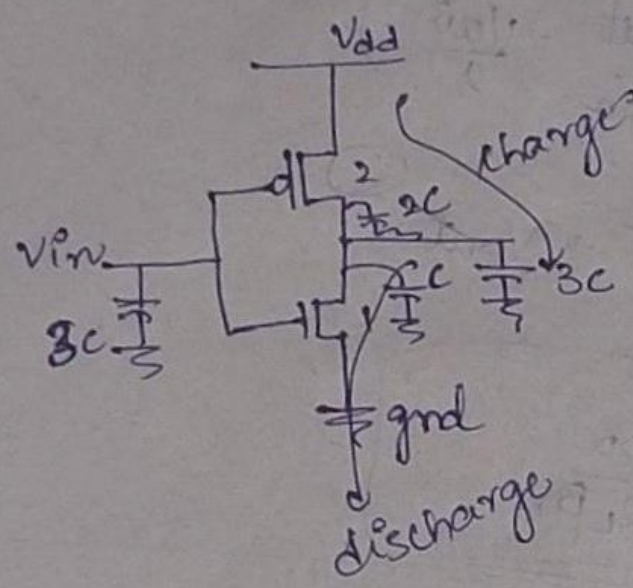
$$R_{ov} = \frac{1}{\frac{\partial I_D}{\partial V_{DS}}} = \frac{1}{\beta_n (V_{gs} - V_{th} - V_{ds})}$$

$$\frac{\partial I_D}{\partial V_{DS}} = \frac{1}{R}$$

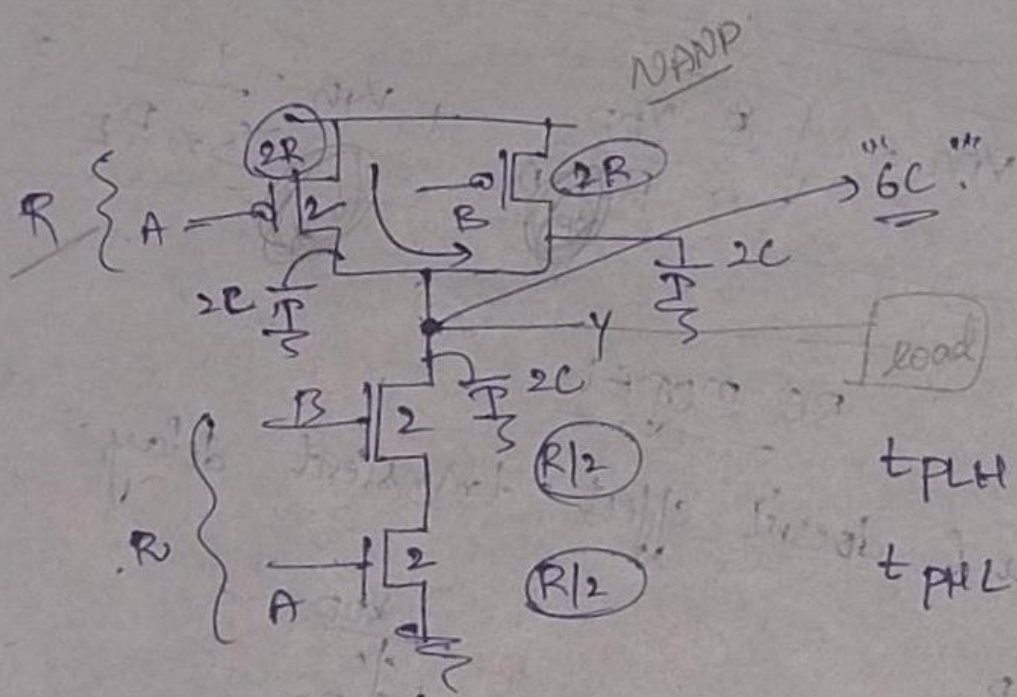
$R_p = 2R_n$
beoz of e^- in NPMOS

Standard
NOT

$\Rightarrow$ $W_L \times 2$ PMOS of W_L NMOS



$$\tau = 3RC$$



$$t_{PLH} = 6RC$$

$$t_{PHL} = RC + 6RC$$

$$= 7RC$$

$$t_{PLH} =$$

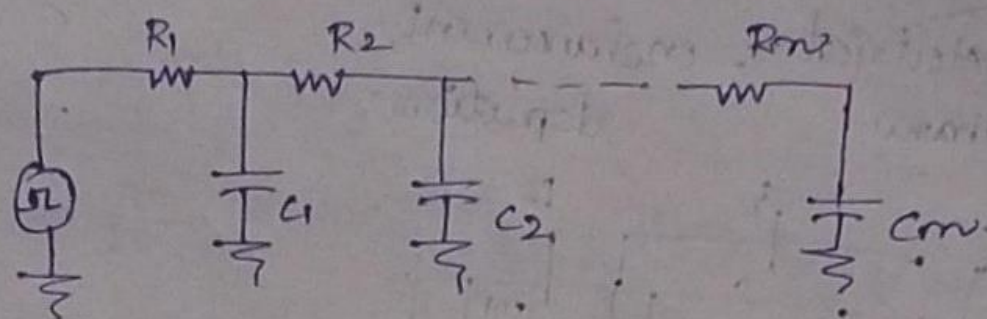
$$\propto \frac{1}{\text{Resistance}}$$

W_i
Aspect
Ratio

$$t_{PHL} = RC + 6RC$$

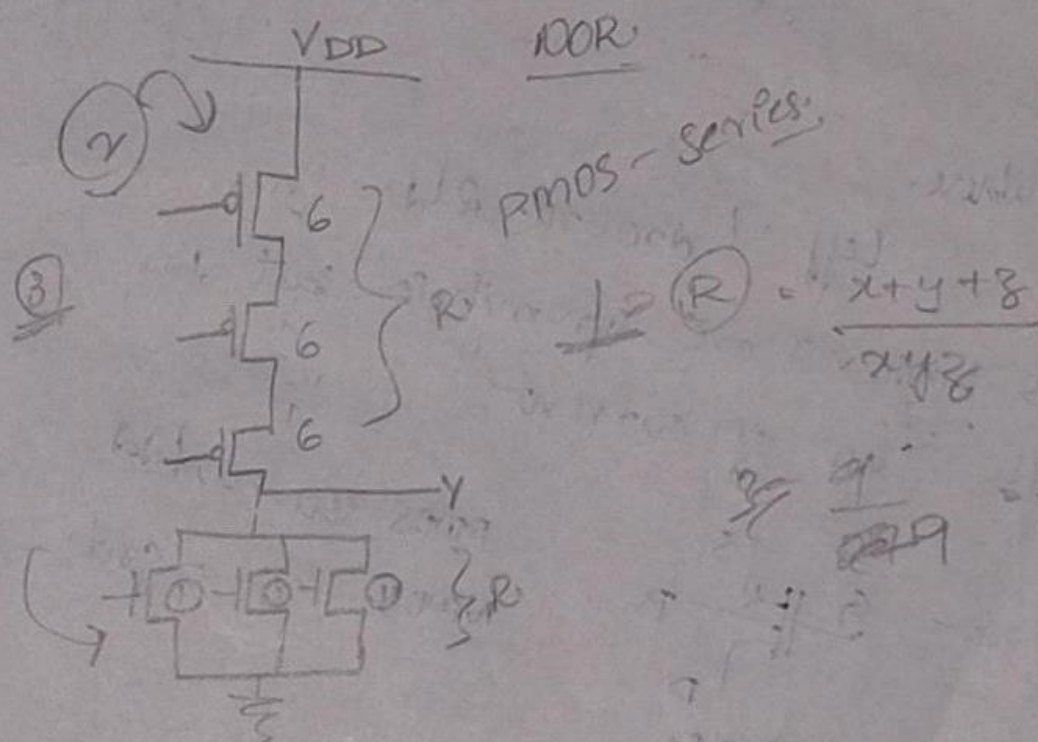
$$= 7RC + 4RC$$

Elmore Delay



$$\text{delay} = \sum_{i=1}^N C_i \sum_{j=1}^i R_j$$

$$= R_1 C_1 + (R_1 + R_2) C_2 + (R_1 + R_2 + R_3) C_3 + \dots$$



Pulse

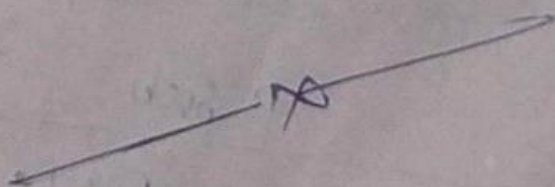
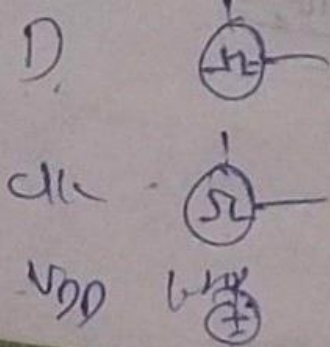
DFF - LTspice

NAND

V_{DD}
V_{SS} - gnd.
V_{1,2} - ip
V_O - op
V₃ - V_{DD}

NOT

V_{DD} - V_{DD}
V_{in} - ip
V_{out} - op
V_{SS} - gnd.



18/2/24

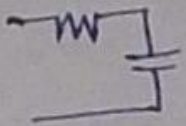
Logical Effort

(d)_{abs} = $d\tau$ → due to process technologies.
↓
absolute delay

⇒ $d = f + p$ → effort delay / stage = parasitic delay

⇒ $f = g \times h$ → electrical effort (op/in) cap
↳ logical effort

$$\tau = 0.69RC$$



for simple NOT

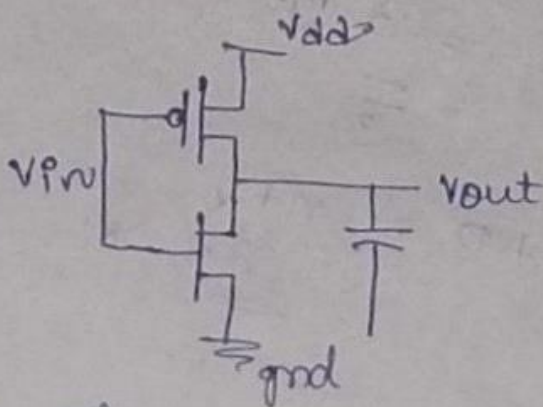
$$\tau = RC$$

but for actual gates $\tau \neq RC$

Inverter - logical effort

delay in NOT - T_{PLH}, T_{PHL} → ON NMOS

↓
ON of PMOS

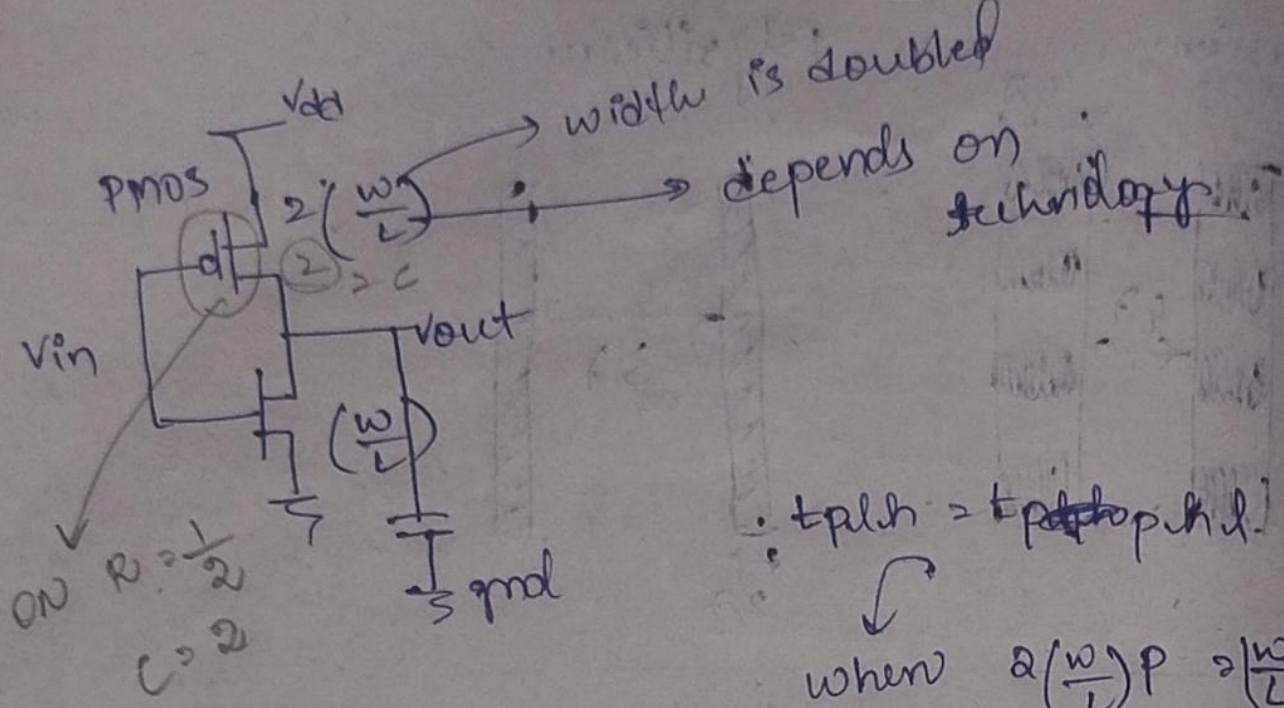


charge - holes → PMOS ON

discharge - e^- through NMOS

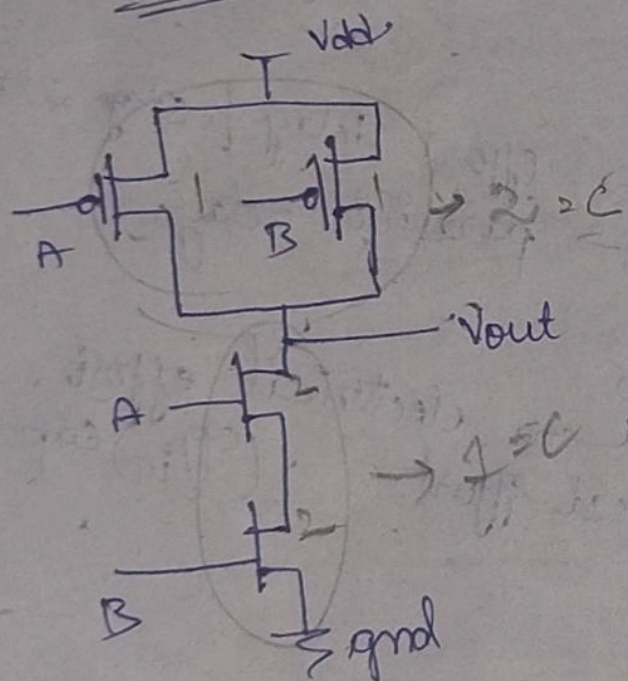
compensate dble pmos size

← $\mu_{e^-} \gg \mu_{holes}$
mobility of $e^- \gg$ mobility of holes



$t_{plh} = t_{poh}$
when $2\left(\frac{w}{L}\right)P = \left(\frac{w}{L}\right)N$
• P-P, N-S
+ P-S, N-P

2 ip NAND



$V_{out} = 1$ either $A/B = 0$

I want $R = \frac{1}{2}$, $C = 2$ in PMOS

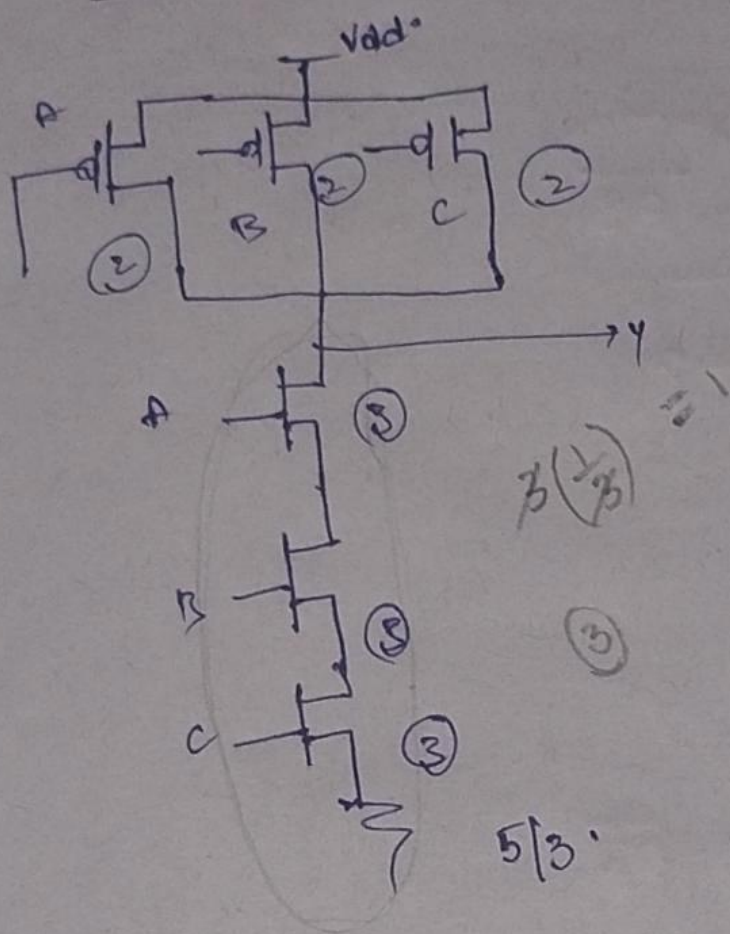
Resistance
series $\rightarrow$ division
parallel $\rightarrow$ same value

logical effort 2ip NAND = $\frac{4}{3}$

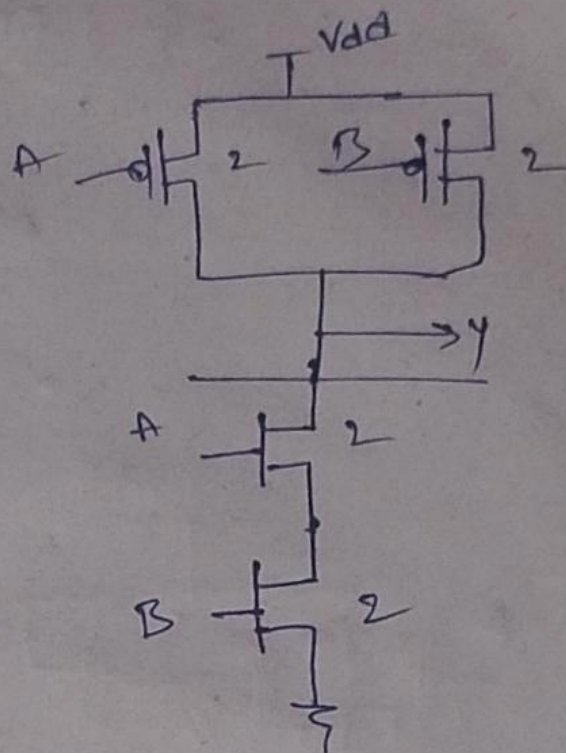
3ip NAND = $\frac{5}{3}$
4ip " = $\frac{6}{3}$
5ip " = $\frac{7}{3}$

nip NAND = $\frac{n+2}{3}$ *

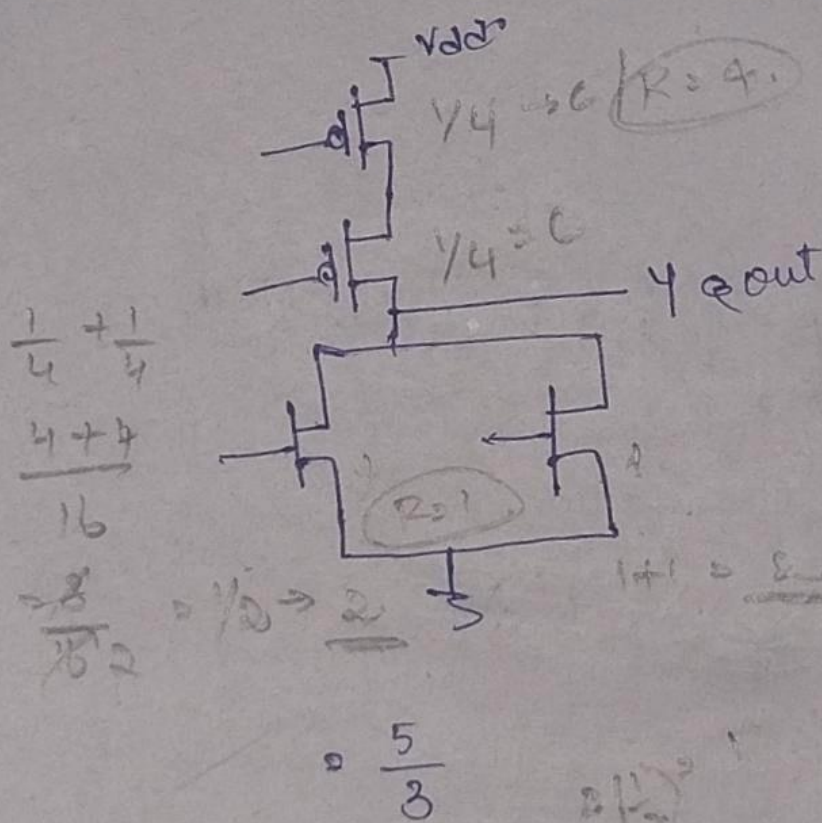
3 ip NAND.



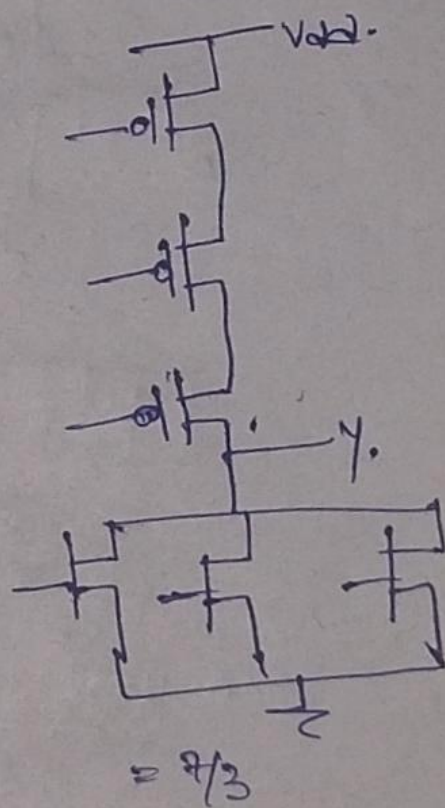
2 ip NAND.



2 ip NOR.



3 ip NOR.



$$n \text{ ip NOR} = \frac{2(n+1)}{3}$$

ip $\uparrow$ α delay $\uparrow$